

University Laboratory Booking & Management System

Complete User Manual, Run Guide & Stakeholder Verification Document

1. System Overview & Architecture

The **University Laboratory Booking & Management System** is a desktop & web-capable application built with strict **Backend/ Frontend folder separation**, **C++17 Qt 6 Widgets**, and a **Python/PySide6 Desktop Application Suite**.

Folder Organization:

```
lab_booking_system/
├── backend/                                # Database schema, C++ OOP
│   ├── domain classes, Python backend engine
│   ├── db/schema.sql                      # SQLite Schema & seed data
│   ├── include/ & src/                   # C++ Header &
│   │   └── Implementation files (User, Student, BookingManager, etc.)
│   └── app/                              # Python SQLite connection
│       ├── manager & BookingManager logic
│       └── frontend/                    # UI Layer (C++ Qt windows &
│           └── PySide6 Desktop GUI)
│               ├── include/ & src/        # C++ Qt UI Window headers &
│               │   └── implementations
│               └── desktop/main.py        # PySide6 Desktop GUI
│                   └── application entry point
├── LabBookingSystem.pro                   # Qt Creator project file
└── README.md                             # System Overview &
    Instructions
```

2. How to Run the Application

You can launch and test the system using any of the following methods:

Method A: PySide6 Desktop GUI Application (Recommended)

Run the following command in terminal or PowerShell:

```
python frontend/desktop/main.py
```

Features native Windows Qt 6 GUI styling with modern dark theme and a **Stakeholder Role Dropdown**.

Method B: C++ Qt Creator Project

1. Launch **Qt Creator**.
2. Open LabBookingSystem.pro.
3. Click **Build & Run** (Ctrl + R).

Method C: Web Application Server

1. Start the HTTP backend server:

```
python backend/app/server.py
```

2. Open your web browser at: `http://localhost:5000`
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3. Stakeholder Login Credentials & Roles Summary

Use the **Select Stakeholder Role** dropdown on the login screen to instantly select any role:

Stakeholder Role	University E-ID	Password	Profile Displayed	Key Permissions
Administrator	ADMIN001	admin123	Dept: Nano Science System Admin	Manage Approvals Queue, Equipment Inventory, Reports
Student	249109	pass123	Index: 249109 BSc Nano	View Labs, Check Slot Availability,

Stakeholder Role	University E-ID	Password	Profile Displayed	Key Permissions
Lecturer	LEC001	pass123	Science Year 2	Request Lab Session
			Emp ID: EMP001 Senior Lecturer Nano Science	Reserve Labs for Class Practicals & Research

4. Stakeholder Testing & User Manual

A. Testing as a Student (249109 / pass123)

1. Login:

- Open the app -> Select **Student (Undergraduate / Postgraduate)** from the dropdown -> Click **Login to Stakeholder Portal**.

2. Verify Student Profile:

- Look at the top header bar. Verify it displays: Welcome, Student A [Student] Index No: 249109 | BSc Nano Science | Year 2.

3. Test Double-Booking Slot Checker:

- Click **Laboratories Directory** tab.
- Select Nano Materials Lab, Date 10/09/2026 (dd/MM/yyyy), From 10:00, To 12:00.
- Click **Check Slot**.
- **Expected Result:** Displays **UNAVAILABLE: Conflict detected! Lab is ALREADY BOOKED for this slot.** (Because an approved booking already exists for this slot).
- Change time slot to From 14:00, To 16:00 -> Click **Check Slot**.
- **Expected Result:** Displays **AVAILABLE: Lab is free for the selected date and time slot.**

4. Submit a Lab Booking Request:

- Click **Booking Portal** tab.
- Form Title will show: **Submit Student Individual / Assignment Lab Session Request.**
- Select Course: NS2001 - Practical Nanotechnology.
- Select Lab: Chemistry Lab.
- Select Date: 15/09/2026 (Notice the date format is strictly dd/MM/yyyy).
- Select Time: 09:00 to 11:00.
- Enter Purpose: Spectroscopy Project Assignment.
- Click **Submit Booking Request**.

- **Expected Result:** Success alert box **Booking #2 submitted successfully and is pending approval.** appears, and the request appears in **My Booking Requests History** with status Pending.
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B. Testing as a Lecturer (LEC001 / pass123)

1. **Login:**
 - Click **Logout** -> Select **Lecturer (Academic Staff)** from the dropdown -> Click **Login**.
 2. **Verify Lecturer Profile:**
 - Header bar displays: Welcome, Dr. Perera [Lecturer / Academic Staff] Employee ID: EMP001 | Senior Lecturer | Nano Science.
 3. **Reserve Class Practical Lab:**
 - Click **Booking Portal** tab.
 - Form Title will show: **Reserve Laboratory for Class Practical / Research.**
 - Select Course Code: NS3010 - High Resolution Spectroscopy.
 - Select Lab: Nano Materials Lab.
 - Select Date: 16/09/2026 (dd/MM/yyyy), Time: 10:00 to 12:00.
 - Enter Purpose: Class Practical Session for 25 Students.
 - Click **Submit Booking Request**.
 - **Expected Result:** Booking request submitted with status Pending.
 4. **Verify Nav Permissions:**
 - Notice that the **Manage Approvals** tab is hidden (preventing students/lecturers from approving their own bookings).
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C. Testing as an Administrator (ADMIN001 / admin123)

1. **Login:**
 - Click **Logout** -> Select **Administrator (Admin)** from the dropdown -> Click **Login**.
2. **Verify Administrator Profile:**
 - Header bar displays: Welcome, Admin [Administrator] Dept: Nano Science | System Admin Privileges.
3. **Verify Dashboard Metrics:**
 - Dashboard card #3 displays **Pending Approvals Queue** badge (e.g. 2 pending requests).
4. **Approve / Reject Requests in Approvals Queue:**
 - Click **Manage Approvals** tab.
 - You will see the pending booking requests submitted by Student A and Dr. Perera.
 - Select a row in the table -> Type remarks: Approved by Lab Admin for Nano Science Practical -> Click **Approve Booking**.

- **Expected Result:** Information popup **Booking approved** appears, and the booking disappears from the pending queue.
5. **Equipment Inventory Management:**
- Click **Equipment Inventory** tab.
 - View inventory items (Microscope, Centrifuge, Spectrophotometer, Analytical Balance).
6. **System Analytics & Reports:**
- Click **System Reports** tab.
 - Change Status Filter dropdown from All to Approved or Pending.
 - **Expected Result:** Table updates dynamically displaying filtered report logs.
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5. Double-Booking Prevention Verification Rules

The core algorithm enforces that **no two bookings for the same laboratory overlap on the same date.**

Conflict Condition: (Existing Start < New End) AND (Existing End > New Start)

Test Matrix:

Lab	Date (dd/MM/yyyy)	Slot Tested	Existing Booking	Expected Result
Nano Materials Lab	10/09/2026	10:00 - 12:00	10:00 - 12:00 (Approved)	UNAVAILABLE (Blocked)
Nano Materials Lab	10/09/2026	11:00 - 13:00	10:00 - 12:00 (Approved)	UNAVAILABLE (Blocked)
Nano Materials Lab	10/09/2026	14:00 - 16:00	None	AVAILABLE (Allowed)